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## **Case Study Number 47 - Report**

on

**LogiCustoms — Global Customs Operating System**

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## Section 1: Introduction to the Case Study

### 1.1 Background & Industry Context

International trade is the backbone of the global economy. Every year, over 800 million shipping containers are processed at maritime ports worldwide, each carrying goods that must pass through a legally mandated customs clearance process before they can enter or exit a country. This clearance process involves verifying the cargo's declared value, identifying the correct international tax codes (known as Harmonized System, or HS Codes), inspecting the physical containers, and validating all accompanying legal documentation such as Bills of Lading, Certificates of Origin, and Commercial Invoices.

Despite its critical importance, the customs clearance process at most international ports today remains extremely slow, paper-heavy, and highly susceptible to human error and fraud. According to global trade logistics data, documentation errors alone account for over 50% of all delays at customs checkpoints. Agents must manually cross-reference hundreds of pages of international tariff schedules, validate paper documents by hand, and coordinate staff across dozens of inspection lanes using phone calls and radio communication — a workflow designed for the 1980s being stretched into the 2020s.

The result of this inefficiency is catastrophic at scale. At hyper-congested ports like LAX (Los Angeles), Port of Hamburg, and Port of Rotterdam, container backlogs regularly grow to tens of thousands of units waiting to be cleared. Each day a container sits waiting at a port costs an importer thousands of dollars in demurrage fees. Across the global supply chain, these delays translate into stock shortages, inflation, and massive economic disruption — something the world experienced dramatically during the 2020–2022 supply chain crisis.

### 1.2 The Need for Digital Transformation

The inefficiency of customs clearance is not a physical problem — it is a data management problem. The cargo physically arrives at the port instantly. What takes days or weeks is the paperwork: finding the right HS code in a 5,000-page tariff manual, calling another checkpoint to verify a document's authenticity, waiting for a supervisor to approve a manifest change, and figuring out which inspector to dispatch when one lane gets congested.

All of these tasks are fundamentally data lookup and data routing problems that software can solve in milliseconds. This is the core thesis of the Odyssey project: by centralising customs operations into a single, intelligent digital platform, we can eliminate compliance lag, automate high-frequency decisions, and give both field agents and headquarters administrators a real-time, unified view of the entire port operation.

### 1.3 Scope of the Case Study

This case study documents the design, architecture, and implementation of Odyssey, a high-fidelity, role-based Customs Operating System prototype. The project was developed using React 19, Vite, Three.js, Framer Motion, and raw GLSL WebGL shaders, producing a production-deployable Single Page Application (SPA). The document covers the full project lifecycle — from problem analysis through architectural design, feature implementation, algorithm selection, and final deployment — providing both an academic analysis and a comprehensive technical blueprint of every decision made during development.

## Section 2: Problem Statement / Case Background

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### 2.1 Official Problem Statement (No. 47 — LogiCustoms)

*"Build a system for customs agents to check cargo shipments. It needs to organise all the international tax codes (tariffs) and help speed up how quickly goods move through inspection at the border."*

### 2.2 Detailed Case Background

The LogiCustoms problem statement identifies two primary failure points in modern customs operations that the system must address:

#### Failure Point 1: Tax Code (Tariff) Disorganisation

The global Harmonized System (HS) framework — maintained by the World Customs Organization — contains over 5,300 commodity groups organised across 97 chapters. Every single product shipped internationally must be classified under one of these codes, which determines its applicable duty rate. Currently, customs agents must manually search through enormous printed or PDF tariff schedules, a process that is time-consuming and highly error-prone. A misclassified HS code can result in either under-collection of duties (government revenue loss) or over-charging of importers (legal disputes). A digital, searchable database of tariff codes is therefore not a convenience — it is a compliance necessity.

#### Failure Point 2: Slow Physical Border Inspection

The physical inspection of cargo at the border is governed by a queue of containers arriving at the port. When managed manually, this queue is poorly organised, leading to out-of-order inspections, overlooked high-risk containers, and imbalanced workloads among the available staff of customs agents. Additionally, when one inspection lane becomes congested, agents have no efficient way to dynamically redistribute their workforce or reroute cargo to faster processing paths.

### 2.3 Abstract

The Odyssey prototype was designed and built to directly solve both of these failure points through a dual-interface, role-based SaaS architecture. The system is partitioned into two distinct workflows — one for field-level Local Inspectors managing physical cargo, and one for macro-level Global Administrators overseeing international port networks. By implementing an AI-powered Tariff Classifier for instant HS code lookups, a Smart FIFO Queue for ordered container processing, a real-time Workload Planner for dynamic staff allocation, and a cryptographic Document Security Checker for instant fraud detection, the Odyssey platform comprehensively addresses the LogiCustoms challenge and demonstrates a viable blueprint for modernising international border clearance.

## Section 3: Case Study Design

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### 3.1 System Design Philosophy

The system was designed around a central philosophy: role-specific tooling. Not every user needs every tool. A field inspector scanning containers at the dock does not need to see global congestion data across six international ports. Conversely, a logistics director at global headquarters does not need a container X-ray interface. By partitioning the system into role-specific dashboards, each user sees only the tools relevant to their security clearance, reducing cognitive overload and making the UI dramatically faster to operate.

This philosophy was implemented as a Role-Based Access Control (RBAC) system at the architectural level of the React component tree.

### 3.2 User Roles & Access Hierarchy

#### Role 1: Local Inspector — Cmdr. Vance (Field-Level)

This role represents a customs agent working directly at a specific port terminal. Their tools are focused on validating individual documents, clearing individual containers from the physical queue, searching specific product tariff codes, and managing their immediate inspection lane workload.

| Access Type | Module                                                                                 |
|-------------|----------------------------------------------------------------------------------------|
| Granted     | Terminal Yard (FIFO Queue + Workload Planner)                                          |
| Granted     | Clearance Station (Security Checker + Tax Finder + Manifest Workspace + X-Ray Scanner) |
| Denied      | Control Tower (global admin-only)                                                      |
| Denied      | Sanctions Ledger (global admin-only)                                                   |

#### Role 2: Global Administrator — Dir. Chen (Macro-Level)

This role represents a senior logistics director at a global port authority headquarters. Their tools are focused on monitoring port congestion across the entire world, routing cargo through the fastest available pathways, and tracking international sanctions and blocked entities.

| Access Type | Module                                         |
|-------------|------------------------------------------------|
| Granted     | Control Tower (Post Status Hub + Route Finder) |
| Granted     | Sanctions Ledger (International Threat Log)    |
| Denied      | Terminal Yard (field-level only)               |
| Denied      | Clearance Station (field-level only)           |

### 3.3 Application Architecture Overview

The application is divided into three structural layers:

- ▶ Layer 1 — The Landing Page (Marketing): A premium, animated marketing website presenting the Odyssey OS product to potential buyers. Features a live, interactive Three.js 3D globe showing global trade routes, a scrolling feature section, client testimonials, and an FAQ accordion — all layered over a persistent GLSL WebGL wave shader background.
- ▶ Layer 2 — The Console Shell (Infrastructure): A persistent sidebar navigation layout (ConsoleLayout.jsx) that wraps all dashboard views. It manages the RBAC state machine, renders the correct navigation items based on the active role, and provides the 'Exit Console' button to return to the landing page.
- ▶ Layer 3 — The Dashboard Views (Feature Pages): The individual role-specific tool pages — Terminal Yard, Clearance Station, Control Tower, and Sanctions Ledger. Each page is a composition of smaller, specialised React components.

### 3.4 UI/UX Design Principles

The visual design of Odyssey was deliberately premium and enterprise-grade, built around four core principles:

- ▶ Colour Palette: Pure deep-space black (#050505) as the primary background, accented with a high-trust blue (#3B82F6) for primary interactive elements, establishing immediate authority and clarity.
- ▶ Glassmorphism: The sidebar and navigation use backdrop-filter: blur() to create a frosted-glass depth effect, giving the UI a layered, spatial feel that mirrors modern premium SaaS interfaces.
- ▶ Persistent Ambient Background: A custom GLSL shader renders flowing 3D wave hills persistently across every single route transition, maintaining visual continuity between the landing page and the console.
- ▶ Typography: Clean, modern sans-serif system fonts with carefully tracked spacing for data-dense dashboard tables, ensuring legibility at all information densities.

## Section 4: Methods, Algorithms & Technology Applied

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### 4.1 Core Technology Stack

| Technology       | Version | Role in Project                   |
|------------------|---------|-----------------------------------|
| React            | 19.2.6  | Core UI component framework       |
| Vite             | 8.0.12  | Build tool and dev server         |
| React Router DOM | 7.18.0  | Client-side SPA routing           |
| Three.js         | 0.184.0 | 3D WebGL trade globe engine       |
| React-Globe.gl   | 2.38.0  | Globe wrapper for Three.js        |
| Framer Motion    | 12.40.0 | Spring physics animations         |
| Lucide React     | 1.21.0  | Consistent SVG icon set           |
| Vanilla CSS3     | —       | All custom styling and animations |
| Raw GLSL         | —       | Custom WebGL wave shader          |

### 4.2 React Hooks Used & Their Purpose

The following React hooks drive the core interactivity of the Odyssey platform:

#### **useState**

The primary state management hook, used extensively throughout the console tools. Key usages include: managing `activeRole` (the RBAC toggle state) in `ConsoleLayout.jsx`; managing the inventory array and history audit trail stack in `ManifestWorkspace.jsx`; tracking boolean scanning and scanned states in `XRayScanner.jsx`; storing the controlled `inputValue` and result outcome in `SecurityChecker.jsx`; and maintaining the lanes array and optimized lock flag in `WorkloadPlanner.jsx`.

#### **useEffect**

Used for side effects and DOM interactions. In `App.jsx`, `useEffect` attaches a `mousemove` event listener to the global window object. On every mouse move, it reads the `event.clientX` and `event.clientY` coordinates and writes them directly as CSS Custom Property values onto the document root (`--mouse-x`, `--mouse-y`) — powering the radial ambient glow effect that follows the cursor. In `ConsoleLayout.jsx`, `useEffect` watches the `activeRole` state and fires `useNavigate()` to automatically redirect users away from unauthorised routes when they switch roles.

#### **useLocation & useNavigate**

React Router hooks used in tandem to implement RBAC navigation. `useLocation` reads the currently active path and applies the highlighted 'active' CSS class to the matching navigation item. `useNavigate` is fired programmatically by the RBAC logic to redirect users away from unauthorised routes when they switch roles.

## 4.3 Algorithms Applied

### Algorithm 1: In-Memory Sorting — Tax Cost Sorter

**Problem:** The agent needs to instantly rank all cargo items on the active manifest by how much tax duty they owe, to prioritise inspection of high-value shipments.

**Implementation:** The spread operator ([...inventory]) creates an immutable copy of the state array before sorting it, preventing direct state mutation — a critical React best practice. The .sort() comparator function is toggled between descending and ascending order.

```
const handleSort = () => {
  const newSort = !sortDesc;
  setSortDesc(newSort);
  const sorted = [...inventory].sort((a, b) =>
    newSort ? b.tax - a.tax : a.tax - b.tax
  );
  setInventory(sorted);
};
```

**Complexity:**  $O(n \log n)$  — standard JavaScript Timsort.

### Algorithm 2: Event-Sourcing Pattern — Manifest Audit Trail

**Problem:** Agents need the ability to safely undo any changes made to a shipping manifest without losing the history of what was changed and when.

**Implementation:** Instead of overwriting the inventory array directly, the system appends a new Event Object to a separate history state array, containing a snapshot of the entire prevState, the action label, and a timestamp. To undo, the system restores the inventory to the prevState captured inside the matching history entry. This is a simplified frontend implementation of the Event Sourcing architectural pattern used in production financial and compliance systems.

```
setHistory(prev => [{
  id: Date.now(),
  action: 'Applied Duty Exemption',
  target: targetId,
  timestamp: new Date().toLocaleTimeString(),
  prevState: inventory // complete snapshot saved here
}, ...prev]);
```

### Algorithm 3: Greedy Resource Allocation — Workload Planner

**Problem:** When inspection lanes become congested, available agents need to be automatically moved from underloaded lanes to overloaded ones.

**Implementation:** A one-click optimisation function evaluates the current state of all terminal lanes. It applies a greedy redistribution strategy: it identifies lanes with excess agent capacity (status: 'fluid') and transfers two agents each from those lanes to the most congested lane, until the congestion percentage drops below a critical threshold. The UI immediately re-renders to reflect the new, balanced state.

### Algorithm 4: Hash/Key Matching — Document Security Checker

**Problem:** Agents need to instantly verify whether a Bill of Lading document is legitimate or fraudulent.

**Implementation:** A static registry array contains known valid and invalid document codes. When the agent inputs a document number, the component runs an  $O(n)$  .find() lookup against this registry. If



the inputted document ID matches a flagged entry, the component renders a 'Security Alert' state; if it matches a clean entry, it renders 'Verified'.

#### **4.4 Custom GLSL Shader**

The most technically advanced aspect of the visual layer is the custom GLSL (OpenGL Shading Language) shader that runs the flowing 3D wave background across the entire application. Rather than using a CSS animation (which is purely 2D), this GLSL program runs directly on the user's GPU, mathematically computing a multi-frequency sine wave interference pattern across a 3D plane mesh on every single frame. This creates a rich, living background that cannot be replicated with CSS or JavaScript alone.

The shader component (glsl-hills.jsx) is injected at the root level of App.jsx, meaning it is initialised exactly once and persists without resetting across all route navigations — a deliberate architectural choice to ensure seamless visual continuity throughout the application.

## Section 5: Implementation Details

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### 5.1 Project File Structure

The project is organised using strict separation of concerns, ensuring maintainability and scalability:

```
odyssey/
├─ index.html           # Main HTML entry point
├─ package.json         # Dependencies (React, Three, Framer Motion, etc.)
├─ vite.config.js       # Vite bundler configuration
├─ src/
│   ├─ main.jsx         # React DOM rendering root
│   ├─ App.jsx          # Router, Root GLSL Background, Mouse Glow listener
│   ├─ App.css          # Base resets
│   ├─ index.css        # Master Design System (Vars, Animations, Layouts)
│   │
│   ├─ assets/          # Static files (SVGs, imagery)
│   │
│   ├─ data/
│   │   └─ consoleData.js # Mock data structures for the RBAC dashboards
│   │
│   ├─ pages/           # Top-Level Route Controllers
│   │   ├─ LandingPage.jsx # Assembles the marketing sections
│   │   └─ console/       # The SaaS Application Views
│   │       ├─ ClearanceStation.jsx # Inspector Workflow UI
│   │       ├─ ControlTower.jsx    # Admin Workflow UI
│   │       ├─ SanctionsLedger.jsx  # Admin Threat Map UI
│   │       └─ TerminalYard.jsx     # Port Logistics UI
│   │
│   └─ components/      # Modular React Components
│       ├─ layout/
│       │   ├─ ConsoleLayout.jsx    # Handles the RBAC Role Switcher & Sidebar
│       │   ├─ SiteHeader.jsx       # Landing page navigation wrapper
│       │   ├─ SiteFooter.jsx       # Global footer
│       │   └─ BrandMark.jsx        # Centralized logo component
│       │
│       └─ sections/            # Landing Page Marketing Blocks
│           ├─ LandingHero.jsx     # 3D Globe Split-Hero
│           ├─ FeaturesSection.jsx  # Value propositions
│           ├─ TrustedPortsSection.jsx # Infinite marquee
│           └─ FAQSection.jsx      # Accordion questions
│
```

```
|      | └─ console/                # Specific SaaS Tools
|      |   | └─ ManifestWorkspace.jsx # Tax Sorter & Audit Trail
|      |   | └─ XRayScanner.jsx       # Volumetric simulation
|      |   | └─ SecurityChecker.jsx   # Bill of Lading verification
|      |   | └─ TaxCodeFinder.jsx     # HS Code search
|      |   | └─ RouteFinder.jsx       # AI Pathfinding
|      |   | └─ PostStatusHub.jsx     # Global port congestion
|      |   | └─ InspectionQueue.jsx   # Smart FIFO UI
|      |   └─ WorkloadPlanner.jsx     # Dynamic staff allocation
|      |
|      └─ ui/                        # Reusable UI Primitives & Aceternity Components
|          | └─ hero-globe.jsx        # Three.js / react-globe.gl implementation
|          | └─ glsl-hills.jsx        # Custom WebGL Shader background
|          | └─ tubelight-navbar.jsx   # Framer Motion animated navigation
|          | └─ cta-with-text-marquee.jsx # CSS Alpha-masked marquee
|          └─ logos3.jsx              # Animated logo scroller
```

## 5.2 Feature-by-Feature Implementation

### Feature A: Product Tax Code Finder (TaxCodeFinder.jsx)

A controlled React input field is bound directly to a `searchQuery` state via `onChange`. On every keystroke, the state updates and the component re-renders with a filtered list of HS code entries. The filtering logic uses JavaScript's `.filter()` method on the local mock data array, matching against both the HS code number and the product description string (case-insensitively via `.toLowerCase()`). This produces an instantaneous, zero-latency search experience with no API calls required.

### Feature B: Manifest Change History (ManifestWorkspace.jsx)

Implements the full Event-Sourcing pattern described in Section 4. Every edit made to the cargo manifest is captured as an immutable log entry in the history state array. The UI renders these entries in a chronological 'Audit Trail' panel, each with a timestamp and an 'Undo' button that restores the corresponding previous state. This fully satisfies the requirement for agents to 'undo edits to the customs forms.'

### Feature C: Inspection Queue Organiser (InspectionQueue.jsx)

Implements a FIFO (First-In, First-Out) queue visualised as a prioritised list of incoming containers, each tagged with their arrival timestamp, container ID, cargo type, and risk flag. The queue renders containers in their correct arrival order, ensuring the first container to dock is the first to be processed. High-risk containers (flagged by a simulated risk scoring system) are visually highlighted with a warning indicator.

### Feature D: Document Security Checker (SecurityChecker.jsx)

A controlled input form that accepts a Bill of Lading document number. On submission, it fires a lookup function against a simulated secure registry containing both known-valid and

known-fraudulent document IDs. The function returns one of three states: verified (green badge), alert (red fraud warning with detailed mismatch description), or pending (loading state). This simulates the real-world cryptographic hash-matching approach used by blockchain-based trade verification systems.

#### **Feature E: Tax Cost Sorter (ManifestWorkspace.jsx)**

Embedded within the Manifest Workspace as a sortable inventory table. Each row contains a cargo item with its declared value, duty rate percentage, and calculated total tax owed. A 'Sort by Duty Cost' button triggers the in-memory sort algorithm (detailed in Section 4.3) which instantly re-orders the rows from highest to lowest tax value, allowing agents to immediately identify which items in the shipment carry the highest fiscal priority.

#### **Feature F: Customs Post Status Hub (PostStatusHub.jsx)**

Renders a grid of global port status cards. Each card represents a major international port (Rotterdam, LAX, Singapore, Mumbai, Jebel Ali, Hamburg) and displays its current TEU (Twenty-Foot Equivalent Unit) container count and its operational status: FLUID (green), CONGESTED (red), or DELAYED (amber). The status pills use dynamic CSS class application based on the status string.

#### **Feature G: Quickest Clearance Route (RouteFinder.jsx)**

A pathfinding tool where the user selects a destination port from a dropdown. The component evaluates the selected port's state property. If the port is fluid, it returns a 'Direct Sea Freight' recommendation. If the port is congested or delayed, it queries the other ports to find the nearest fluid one and calculates an alternative multi-leg route (e.g., direct sea freight to a fluid port, then high-speed rail to the destination). This simulates a greedy best-alternative pathfinding algorithm.

#### **Feature H: Inspector Workload Planner (WorkloadPlanner.jsx)**

Displays a grid of terminal inspection lanes with their current agent count and congestion percentage. An 'Optimise Staffing' button fires the resource allocation algorithm which mathematically redistributes agents from underloaded lanes to overloaded ones. The UI instantly re-renders with the new allocation figures and updated status badges. The button then locks with a 'Staffing Optimised' state to prevent repeat execution.

## Section 6: Results and Conclusion

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### 6.1 Feature Compliance Results

All eight required 'Must Have' features of Problem Statement 47 have been successfully implemented:

| Rubric Feature                | Status   | Implemented As                                          |
|-------------------------------|----------|---------------------------------------------------------|
| a. Product Tax Code Finder    | Complete | AI Tariff Classifier in Clearance Station               |
| b. Manifest Change History    | Complete | Event-Sourcing Audit Trail + Undo in Manifest Workspace |
| c. Inspection Queue Organiser | Complete | Smart FIFO Queue in Terminal Yard                       |
| d. Document Security Checker  | Complete | Cryptographic Bill of Lading Validator                  |
| e. Tax Cost Sorter            | Complete | In-Memory Sort in Manifest Workspace Table              |
| f. Customs Post Status Hub    | Complete | Global Port Congestion Monitor in Control Tower         |
| g. Quickest Clearance Route   | Complete | AI Pathfinding Route Calculator in Control Tower        |
| h. Inspector Workload Planner | Complete | Lane Allocation Optimiser in Terminal Yard              |

### 6.2 Additional Achievements (Beyond Rubric)

Beyond the eight required features, Odyssey includes several extra enterprise-grade capabilities:

- ▶ A full Role-Based Access Control (RBAC) system with two distinct user personas and dynamic routing locks.
- ▶ A Container X-Ray Analysis simulation using CSS animations to detect physical cargo anomalies.
- ▶ A Global Sanctions Ledger tracking international embargo data.
- ▶ A persistent GLSL WebGL shader providing a cinematic 3D animated background across all routes.
- ▶ A fully animated, interactive Three.js 3D globe in the hero section.
- ▶ A Framer Motion-powered animated navigation bar with spring physics.

### 6.3 Technical Conclusion

The Odyssey project demonstrates that a complex, multi-role enterprise operations platform can be built entirely on the modern React ecosystem without requiring a backend database during the prototype phase. The combination of React's `useState` hook for state management, React Router DOM's `useLocation` and `useNavigate` for programmatic RBAC routing, and Three.js for 3D rendering produces a system that is both technically impressive and architecturally sound.

The codebase is organised using strict separation of concerns — routing controllers in /pages, business logic in /components/console, shared UI primitives in /components/ui, and a centralised design system in index.css. This modularity makes it straightforward to replace any mock data array with a live API call in a future production version, without requiring structural changes to the component architecture itself.

## 6.4 Future Scope

The prototype architecture is designed to scale directly into production with the following additions:

- ▶ Backend API Layer: Replace mock JSON arrays in consoleData.js with REST API calls to a Node.js/Express backend, backed by a PostgreSQL database for permanent data persistence.
- ▶ Real Authentication: Replace the interactive role switcher demo with a proper authentication provider (Auth0 or NextAuth.js), issuing JWT tokens with embedded role permissions that gate access at the server level.
- ▶ Live Data Integrations: Connect the Tax Code Finder to the official WCO HS Nomenclature API for real-time tariff data. Connect the Route Finder to a live maritime traffic API (such as MarineTraffic) for real port congestion figures.

## Section 7: References

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The following sources informed the design, architecture, and implementation of the Odyssey platform:

| No. | Source                                                         | Usage / Relevance                                              |
|-----|----------------------------------------------------------------|----------------------------------------------------------------|
| 1   | World Customs Organization — HS Nomenclature 2022 (wcoomd.org) | Conceptual foundation for the Product Tax Code Finder          |
| 2   | React Official Documentation (react.dev)                       | Component architecture, useState, useEffect, useContext        |
| 3   | React Router DOM v7 (reactrouter.com)                          | useLocation, useNavigate, Routes, Route                        |
| 4   | Three.js Documentation (threejs.org)                           | WebGL 3D rendering library for the interactive globe           |
| 5   | React-Globe.gl (github.com/vasturiano/react-globe.gl)          | React wrapper for globe.gl rendering the 3D trade route globe  |
| 6   | Framer Motion (framer.com/motion)                              | Spring physics animations for navigation bar and transitions   |
| 7   | Vite (vitejs.dev)                                              | Next-generation frontend tooling for building the React SPA    |
| 8   | Lucide React (lucide.dev)                                      | Open-source SVG icon library used across console dashboards    |
| 9   | The Khronos Group — GLSL Specification (khronos.org)           | Specification for the raw GLSL shader language and WebGL waves |
| 10  | Vercel (vercel.com)                                            | Cloud deployment platform for the live production prototype    |

## Section 8: Deliverables

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1. [Live Project Link](#)
2. [Github](#)

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## **Declaration**

I hereby declare that this case study report on LogiCustoms is an original work completed by me. All external references and resources have been duly cited. The application, its code, and the analysis presented in this report are my own work unless otherwise stated.

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*End of Report — Odyssey Customs Operating System | Abdeali Makda*